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News-Sentiment Tilted Top-20 FFMC Index

An AI/NLP risk engine driving a free-float market-cap index

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PROBLEM & APPROACH

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Can news make a free-float index more risk-aware without turning it into a trading strategy?

THE PROBLEM

- **Size-Only Weighting**
  Free-float market-cap weights depend on size alone; news never enters them
- **Unstructured News Signals**
  Headlines and tweets are rich but unstructured, hard to use in a systematic, auditable way
- **News-based weighting**
  Goal: turn news into a transparent, rule-based weight signal that keeps the index close to its free-float base

OUR APPROACH

1.Read the news



An NLP engine scores each headline or tweet: sentiment (−1 to +1), impact (1–10), news type, confidence

2.Select by size



Top 20 of the S&P 100 by free-float market cap, reviewed quarterly with a 20/25 buffer; news never changes membership

3.Tilt weights daily



Positive, confident news raises a weight, negative news lowers it, inside caps that keep the index near free-float weights

101
S&P 100 symbols (current members)

20
stocks held at all times

64,028
headlines scored out-of-sample

2016-20
daily backtest, Jan 2016 to Jun 2020

SYSTEM DESIGN

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Simple, auditable models; every score is out-of-sample and every weight change is rule-bound

NEWS ENGINE

NEWS FEED
Headlines and tweets, 2016–2020 (64,028 headlines)

→

MASKED TEXT
Company names hidden before scoring

→

SENTIMENT MODEL
TF-IDF + logistic regression gives S, impact, type, confidence C

→ S, C

INDEX ENGINE

MARKET DATA
yfinance: prices, share counts, splits, free-float factors

→

ELIGIBILITY AND FFMC
FFMC = Price × Shares × Free-float, checked at each review date

→

QUARTERLY SELECTION
Top 20/25 buffer, gives base weights

→ w_FFMC

DAILY TILT + RISK CONTROLS

$$w = w_{FFMC} \times e^{\alpha \cdot S \cdot C}$$

- **α = 0.30, neutral zone |S| < 0.15**
- **Max weight 12%, tilt ±25%**
- **Max 1 pp change a day**

↓

DAILY WEIGHTS AND BACKTEST
10 bp cost on weight traded; metrics vs parent index

DATA FLOW RULES

TIMING
News dated day d sets weights at the close of day d+1, the only lag

→

NO LOOK-AHEAD
Model retrained yearly on past data only; labels come from market moves

→

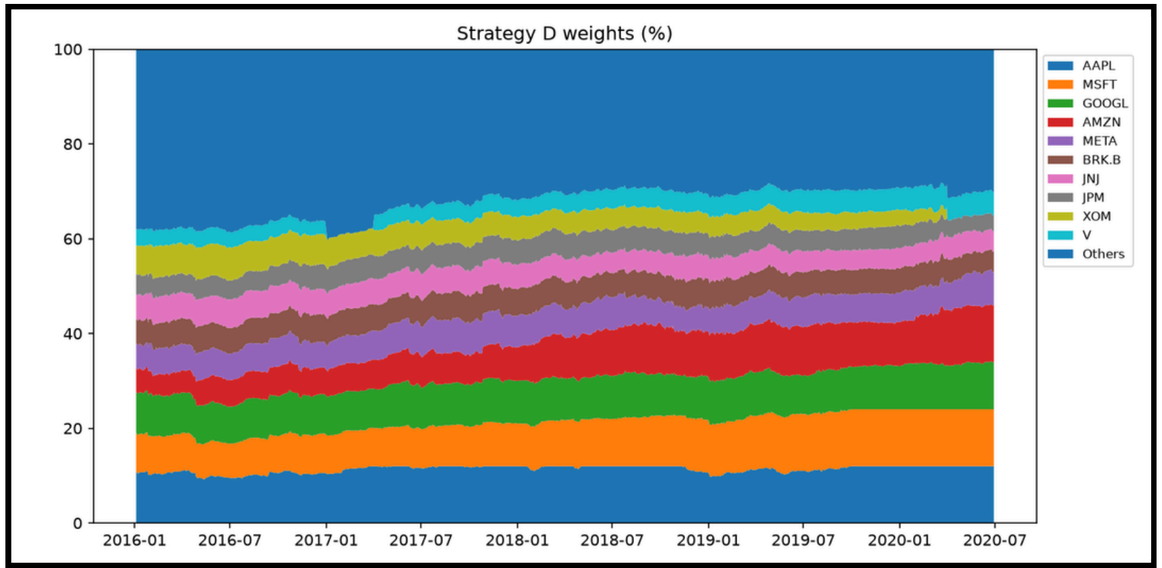
SIZE DECIDES MEMBERSHIP
News never adds or removes a stock

IMPLEMENTATION HIGHLIGHTS

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Two engines run side by side and meet in the daily tilt: size decides membership, news decides weights

Module	Tech choice	Why
Sentiment engine	TF-IDF + logistic regression on masked headlines; labelled by the stock's price move after the headline; retrained every year on past data only	No human labelling and no look-ahead: all 64,028 headlines are scored out-of-sample; probabilities give S = P(pos) − P(neg)
Confidence and decay	C = certainty × independence × agreement × source quality; news half-life 3 trading days	One weak headline moves a weight less than five agreeing ones; old news fades smoothly
Index engine	FFMC ranking, quarterly review, 20/25 buffer	The buffer stops stocks near rank 20 flipping in and out: low turnover
Weight engine	Tilt α = 0.30, neutral zone, 12% cap, ±25% band, 1 pp daily limit	Keeps the index close to free-float weights and limits trading
Data layer	yfinance; 438 splits adjusted, bad points removed, real mergers kept	Clean share counts give a reliable FFMC



Strategy D weights: top names stay at or below the 12% cap

12.00%
max single weight in D (naive tilt B: 16.28%)

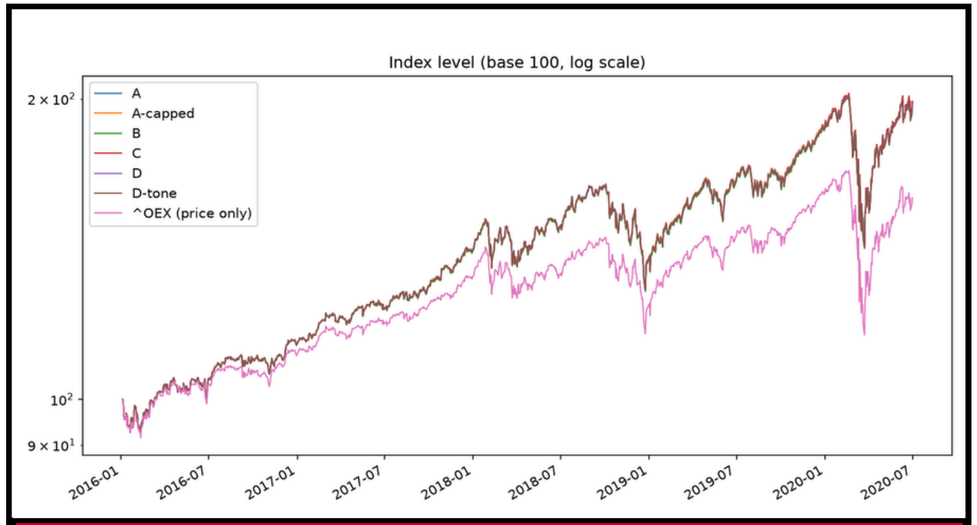
0.89%
average tilt of D from the free-float base weights

KEY RESULTS

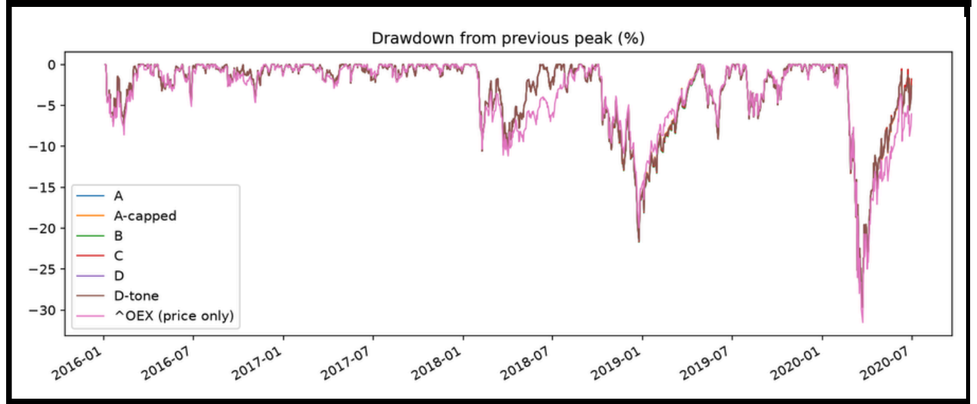
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News is priced in by the next day, so the tilted index (D) tracks the plain free-float index (A)

Jan 2016 – Jun 2020	Parent ^OEX	A: FFMC Top 20	A-capped	B: naive tilt	C: + confidence	D: + risk controls
Total return	59.19%	98.73%	97.62%	97.85%	98.81%	**96.42%**
CAGR	10.93%	16.55%	16.41%	16.44%	16.56%	**16.25%**
Sharpe ratio	0.57	0.8	0.79	0.79	0.8	**0.79**
Max drawdown	−31.53%	−29.70%	−29.71%	−29.60%	−29.64%	**−29.64%**
Tracking error vs ^OEX	–	3.76%	3.68%	3.78%	3.77%	**3.67%**
Tracking error vs A	3.76%	–	0.23%	0.10%	0.06%	**0.31%**
Annual turnover	–	19.02%	20.15%	99.65%	41.45%	**50.37%**
Transaction costs	–	0.17%	0.18%	0.89%	0.37%	**0.45%**



Index levels (log scale): all versions track closely; the tilt changes little



Similar drawdowns; COVID window (19 Feb – 30 Apr 2020): A −9.13% vs D −9.07%

Efficiency vs the naive tilt (B → D)

- Turnover 99.65% → 50.37% and costs 0.89% → 0.45%: about half
- Max single weight 16.28% → 12.00%: the cap holds every day
- A 10× stronger tilt (α = 3) triples costs (0.45% → 1.40%) with no return gain

^OEX is price-only (no dividends); A–D include dividends. A-D includes dividends. Tracking error vs parent vs ACosts: 10 bp × weight traded.

DOMAIN IMPACT

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A safe, auditable way to bring news into index construction and risk monitoring

INDEX AND BENCHMARK PROVIDERS

Rule-based, news-aware weights while membership stays untouched by news. Caps keep the index within an average tilt of 0.89% of its free-float base.

0.31%
tracking error of D vs the plain FFMC index (A)

RISK MONITORING

Every stock-day gets sentiment, impact (1–10), news type and confidence: a structured news-risk feed that can sit beside the index.

50.37%
one-way annual turnover (naive tilt B: 99.65%)

TESTING WHAT NEWS IS WORTH

Out-of-sample scores are right 70% of the time on the day of the news but ~50% the next day (Fama–MacBeth t-stat 0.99): news is priced in by day 1.

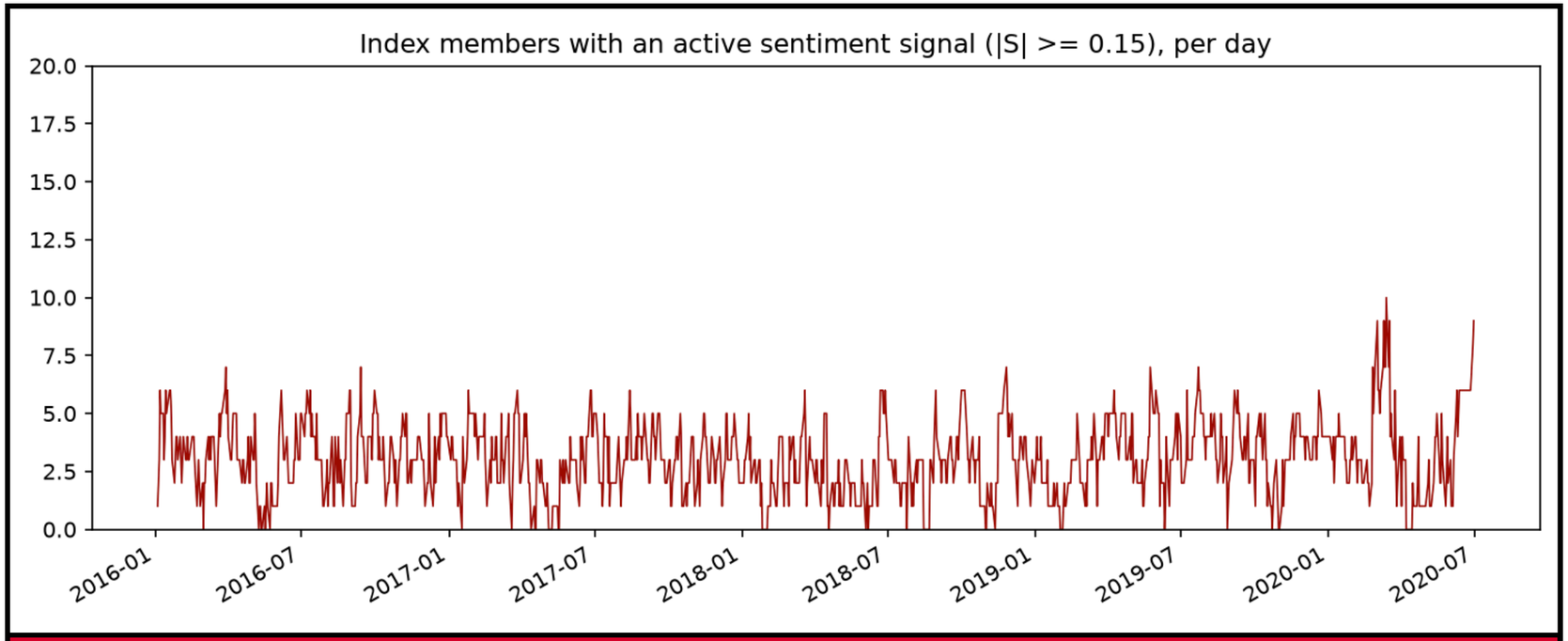
−29.64%
in line with plain FFMC A (−29.70%); parent ^OEX −31.53%

ROBUST AND SAFE, BUT REBALANCING AT THE CLOSE CAN'T CAPTURE THE FIRST-DAY MOVE

LIMITATIONS AND NEXT STEPS

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Timing and coverage limit the signal; both are addressable



Members with an active signal (|S| ≥ 0.15) per day: many never tilt, with no news in the dataset for MSFT, BRK.B, CVX

NEXT STEPS

Intraday news + intraday rebalancing to capture the first-day move

Top-100 universe and a fuller news feed so more stocks tilt

ASSUMPTIONS AND GAPS

Rebalancing Lag
Close-of-day rebalancing misses the first-day move, as news is priced in by Day 1

News Coverage Gaps
Coverage: some stocks never tilt because the dataset has no news for them; Ex: MSFT, CVX

Static Free-Float Factors
Free-float factors are today's values held constant; BRK.B at 0.85 is an estimate

Survivorship Bias
Today's S&P 100 members are used for 2016–2020, so survivorship bias flatters every strategy

Benchmark Limitation
^OEX is price-only, about 2% a year easier to beat than the dividend-inclusive versions